

The homogeneous pair corrector in the singular particle domain

HOCF–RIESZ, Round 008

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Scope and status. This is a synthesis of the submitted THM–028 construction. Its complete proof is the immutable Markdown memorandum; the canonical theorem ledger records the separate independent audit dispositions. The assertion concerns fixed particle number and the exact singular Itô identity. It supplies no uniform fluctuation estimate.

The precise domain assertion

Use the unit Haar torus, $K = -\nabla g$, and the coefficient-one periodic Riesz potential. Fix $d \geq 3$, $0 < s \leq d-2$, $N \geq 2$, $T < \infty$, and $0 \leq \nu \leq \nu_* < \infty$. There is no external drift, and the reference density is one. A smooth real terminal test h determines the actual homogeneous backward test

$$f_t(x) = \widehat{h}(0) + \sum_{k \neq 0} \widehat{h}(k) e^{-(T-t)[4\pi^2 \nu |k|^2 + 4\pi^2 c_{d,s} |k|^{s+2-d}]} e^{2\pi i k \cdot x}, \quad c_{d,s} = \frac{\pi^{s-d/2} \Gamma((d-s)/2)}{\Gamma(s/2)}.$$

All its fixed-order spatial and time derivatives are bounded uniformly on the specified diffusivity interval. Set $E = \{(x, y) : x \neq y\}$ and

$$\begin{aligned} Bv &= K(x-y) \cdot (\nabla_x - \nabla_y)v, & G &= \nu \Delta_{x,y} + B/N, \\ J_t(x, y) &= K(x-y) \cdot (\nabla f_t(x) - \nabla f_t(y)), & R &= R_x + R_y. \end{aligned}$$

The exact homogeneous response is

$$R_x v(x, y) = - \int v(x+w, y) D(dw), \quad D = \operatorname{div} K = \begin{cases} s(d-2-s)g_{s+2}(w) dw, & s < d-2, \\ (d-2)|\mathbb{S}^{d-1}|(\delta_0 - dw), & s = d-2. \end{cases}$$

The second slot is retained by exchange. Let Φ be the previously constructed unique bounded Borel terminal-zero full inverse, with source J and both responses. It is not replaced by a transport diagnostic or the raw potential.

For every $1 < q_1 < d/2$ and $q_1 + 1 < q_2 < d$, the submitted domain assertion is

$$\sup_{0 \leq t \leq T} \left(\|\Phi_t\|_\infty + \sup_E \frac{|\nabla \Phi_t|}{w_{q_1}} + \sup_E \frac{|D^2 \Phi_t|}{w_{q_2}} + \sup_E \frac{|\partial_t \Phi_t|}{w_s} \right) < \infty.$$

Here $w_a(x-y)$ is a fixed positive smooth weight off zero, equal to $|x-y|^{-a}$ near zero and bounded elsewhere. The indicated derivatives are jointly continuous off the diagonal, with one-sided time derivatives at the endpoints. Constants can depend severely on N , but are uniform in the chosen ν . The same representative belongs to global Haar $H^1 \cap W^{2,1}$; $\partial_t \Phi$ and $B\Phi$ belong uniformly in time to Haar L^1 .

Higher derivatives require their own occupation bounds

The local force is $K(z) = s|z|^{-s-2}z + k(z)$ with k smooth and odd. For the pair flow, the largest singular drift-Jacobian eigenvalue is $c(z) = 2s|z|^{-s-2}/N$; the radial eigenvalue is $-(s+1)c(z)$. After a fixed cutoff, a

finite L bounds the smooth remainder, and the first variation satisfies $|A_t| \leq e^{Lt+I_t}$, where $I_t = \int_0^t c(Z_a) da$. For every moment order $m \geq 0$ and weight exponent $\alpha > m$,

$$(G + mc)w_\alpha \leq C_{\alpha,m}w_\alpha - b_{\alpha,m}\chi|z|^{-\alpha-s-2}, \quad b_{\alpha,m} = \frac{s(\alpha-m)}{N} > 0.$$

The positive term $2\nu\alpha(\alpha+2-d)_+|z|^{-2}$ is absorbed by the more singular repulsion, including dimension three. The complete memorandum gives the maximization and all annular constants. Smooth stopping and nonnegative Fatou passage yield weighted terminal and occupation estimates. A maximal supermartingale tail, applied at larger moments, supplies every finite moment on compact sets of initial states.

The expectation derivative is not inferred from pointwise noncollision. Truncated maximal inverse-distance functions of the initial state are locally Lipschitz. Their high Sobolev moments, an explicitly proved local Morrey bound in dimension $2d$, and monotone convergence yield a common noncolliding flow on each compact initial set. First and second chain rules then have moments above one. Their pathwise line identities, Fubini and uniform integrability justify both expectation derivatives and their local continuity.

The second variation is

$$H_t = \int_0^t A_{t,r} D^2 F(Z_r)[A_r, A_r] dr, \quad F = N^{-1}(K, -K).$$

Condition at the intermediate time r to bound the future first variation against w_{q_1} . The remaining occupation has moment order two and source power $s+3+q_1$. The exact inequality

$$s+3+q_1 < q_2+s+2, \quad q_2 > 2,$$

allows the q_2 barrier to control it. This proves the weighted Hessian estimate for terminal data and, separately, the integrated singular source. It does not import a second-derivative conclusion from the first-gradient theorem.

Both homogeneous responses preserve these weighted derivatives. The needed convolution estimate is

$$\int w_a(w)w_q(w+z)dw \leq C_{a,q}w_q(z), \quad a, q < d,$$

with the two singular neighborhoods treated separately. No condition $a+q < d$ is imposed. At Coulomb the atom and compensation replace an invalid power-density argument. A pointwise factorial Volterra series constructs the weighted solution and bounded Borel uniqueness identifies it with Φ .

Time differentiation and the internal-drift cancellation

A separate weighted Borel space with norm $\sup_E |v|/w_s$ supports the base semigroup and the bounded response. An absolutely convergent pointwise Dyson series defines the full autonomous evolution $\mathcal{T}_r$. Local continuity on the specific continuous weighted inputs follows from common-flow moments and uniform integrability. Strong continuity on the entire weighted supremum space is neither needed nor asserted.

The source is twice differentiable in time in that weighted norm. Therefore ordinary differentiation of the time integral gives

$$\begin{aligned} \Phi_t &= \int_0^{T-t} \mathcal{T}_r J_{t+r} dr, \\ \partial_t \Phi_t &= -\mathcal{T}_{T-t} J_T + \int_0^{T-t} \mathcal{T}_r \partial_t J_{t+r} dr. \end{aligned}$$

This differentiates the input and endpoint, not a semigroup generator on an unjustified domain. Comparing the established base-pair martingale with localized classical Itô gives

$$\partial_t \Phi + \nu \Delta_{x,y} \Phi + B\Phi/N + R\Phi = -J \quad \text{on } [0, T] \times E.$$

Consequently

$$B\Phi = -N(J + \partial_t \Phi + \nu \Delta_{x,y} \Phi + R\Phi).$$

Each right-hand term has an integrable majorant: w_s, w_s, w_{q_2} and a constant. This structural identity supplies the L^1 internal drift. The crude force-times-gradient majorant is generally not integrable at Coulomb and is not used for this repeated-pair term.

Deleting a diagonal tube in weak integration by parts gives boundary errors of orders ε^{d-1} and ε^{d-1-q_1} . They vanish because Φ is bounded and $q_1 < d-1$. The conditions $2q_1 < d$ and $q_2 < d$ give H^1 and $W^{2,1}$. Moving-tube estimates justify differentiated background integrals, including their continuous representatives and the vanishing integrated background Laplacian. No diagonal trace is assigned.

The actual finite-particle identity

Let $\eta = N^{-1} \sum_i \delta_{X_i}$ and $\rho = \eta - dx$. For a symmetric pair kernel, write $q(x) = \int \Phi(x, y) dy$ and $r = \int \Phi dx dy$. Then

$$P[\Phi] = \frac{1}{2N^2} \sum_{i \neq j} \Phi(X_i, X_j) - \frac{1}{N} \sum_i q(X_i) + \frac{1}{2} r.$$

For a symmetric triple F , the exact ordered deleted-label statistic is

$$U_3[F] = \frac{1}{N^3} \sum_{i,j,k \text{ distinct}} F(X_i, X_j, X_k) - \frac{3}{N^2} \sum_{i \neq j} F_1(X_i, X_j) + \frac{3}{N} \sum_i F_2(X_i) - F_0,$$

Here F_1 and F_2 integrate one and two Haar coordinates, respectively, and F_0 is the full three-coordinate Haar integral. At $N = 2$ the first sum is empty, but the background terms remain. Define

$$C\Phi = \text{Sym}_3[K(x-z) \cdot \nabla_x \Phi(x, y)], \quad (B\Phi)_\mu(x) = \int B\Phi(x, y) dy,$$

where Sym_3 averages all six permutations. Explicit finite sums separate the repeated internal pair from three distinct labels before any estimate. They give

$$dP_t[\Phi_t] = \left\{ -P_t[J_t] + U_{3,t}[C\Phi_t] + \frac{1}{N} \rho_t[(B\Phi_t)_\mu] + \frac{1}{2N} \int B\Phi_t dx dy \right\} dt + dM_t^2.$$

The gradient and its noise are exactly

$$\begin{aligned} \nabla_i P[\Phi] &= \frac{1}{N^2} \sum_{j \neq i} \nabla_1 \Phi(X_i, X_j) - \frac{1}{N} \int \nabla_1 \Phi(X_i, y) dy, \\ dM_t^2 &= \sqrt{2\nu} \sum_i \nabla_i P_t[\Phi_t](X_t) \cdot dW_i(t), \quad d\langle M^2 \rangle_t = 2\nu \sum_i |\nabla_i P_t[\Phi_t](X_t)|^2 dt. \end{aligned}$$

There is no extra thermal trace: distinct particle labels have independent Brownian noises, and integrated Haar Laplacians vanish.

For a bounded initial N -body density F_0 , independent of the noises, the established actual-law bound is $\|F_t\|_\infty \leq \|F_0\|_\infty e^{(N-1)\kappa T}$, where $D \geq -\kappa dx$. Triple terms are Haar integrable because, conditional on one coordinate, the force and gradient use two independent integration variables with separately integrable powers. Repeated pairs instead use the structural $B\Phi$ bound. All drift terms have finite expected absolute time integrals, and H^1 supplies a finite expected bracket at fixed N . The stopped drift passes in L^1 and the stopped stochastic integral in L^2 ; the latter is a true square-integrable martingale. This is a direct collision localization argument, not a claim of heat-regularized source convergence.

Remaining assertion. Uniform smallness of evolved residuals and self/cross brackets is still open. The exponential density cost and fixed- N derivative constants cannot provide it. The next reference-Haar energy task retains the actual two- and three-particle correlation transfer as a separate obligation. The old energy-floor condition, full microscopic subcriticality and finite positive critical coupling remain distinct.